Proof of Theorem

The n_1 indistinguishable objects of type 1 can be placed among the n positions in $C(n,n_1)=n!/n_1!(n-n_1)!$ ways.

The n_2 indistinguishable objects of type 2 can be placed among the $(n-n_1)$ positions in $C(n-n_1,n_2)=(n-n_1)!/n_2!\ (n-n_1-n_2)!$ ways.

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The n_{k-1} indistinguishable objects of type $k-1$ can be placed among the $(n-n_1-\ldots-n_{k-2})$ positions in $C(n-n_1-\ldots-n_{k-2},\ n_{k-1})= (n-n_1-\ldots-n_{k-2})!/\ n_{k-1}!\ (n-n_1-n_2\ \ldots-n_{k-2}-n_{k-1})!$ ways.

The n_k indistinguishable objects of type k can be placed among the $(n-n_1-\ldots-n_{k-1})$ positions in $C(n-n_1-\ldots-n_{k-1},\ n_k)= (n-n_1-\ldots-n_{k-1})!/\ n_k!\ (n-n_1-n_2\ \ldots-n_{k-1}-n_k)!$ ways.

$$\frac{n!}{n_1!\,(n-n_1)!}\cdot\frac{(n-n_1)!}{n_2!\,(n-n_1-n_2)!}\cdot\ldots\cdot\frac{(n-n_1\ldots-n_{k-2})!}{n_{k-1}!\,(n-n_1\ldots-n_{k-1})!}\cdot\frac{(n-n_1\ldots-n_{k-1})!}{n_k!\,(n-n_1\ldots-n_k)!}=\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!\,0!}=\frac{n!}{n_1!\,n_2!\ldots n_{k-1}!\,n_k!}$$